PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 12 – Re-evaluate and Adapt the Parking System

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1. Task12.md

Aim

To enhance the existing Campus Smart-Parking System according to new requirements.

New Requirements

1. Add e-bikes with a flat parking fee.
2. Delivery vans pay double the car fee between 9 AM and 5 PM.
3. Provide visitors with a one-day parking pass.
4. Extend the parking facility to five floors with different slot counts.

Impact Analysis

- New vehicle types require additional classes.
- Special fee rules require changes in fee calculation.
- Visitor passes require pass management.
- Five floors require an expandable parking structure.

Design Decision

The existing system is **MODIFIED** instead of completely redesigned. Existing classes and parking allocation methods are reused with additional features.

Justification

Modification reduces development cost and time. It also maintains existing functionality and supports future expansion.

Conclusion

The enhanced system provides flexible parking management with new vehicle types, special fees, visitor passes and five-floor parking facilities.

2. Java Code

```
public class Task12 {

    public static void main(String[] args) {

        int[] floors = {4, 5, 4, 6, 5};

        int total = 0;

        System.out.println("Campus Smart Parking System");

        for (int i = 0; i < floors.length; i++) {

            System.out.println("Floor " + (i + 1) +

                               " Slots: " + floors[i]);

            total += floors[i];

        }

        System.out.println("Total Slots: " + total);

        System.out.println("\nParking Fees:");

        System.out.println("E-Bike: Rs.15");

        System.out.println("Car: Rs.30 per hour");

        System.out.println("Delivery Van (9 AM-5 PM): Rs.60 per hour");

        System.out.println("Visitor One-Day Pass: Rs.100");

    }

}
```

3. Output:

```
Campus Smart Parking System

Floor 1 Slots: 4
Floor 2 Slots: 5
Floor 3 Slots: 4
Floor 4 Slots: 6
Floor 5 Slots: 5

Total Slots: 24

Parking Fees:
E-Bike: Rs.15
Car: Rs.30 per hour
Delivery Van (9 AM-5 PM): Rs.60 per hour
Visitor One-Day Pass: Rs.100
```